

My research interest lies in building next-generation computer systems, especially for machine learning workloads. This aspiration originated the first time when I built an end-to-end training pipeline for Graph Neural Network (GNN) at AWS Shanghai AI Lab, and was further fostered during my time at NYU Systems Group working on Large Language Model (LLM) and AI pipelines. Having my research skills and taste shaped by my experience in both industrial research labs (AWS Shanghai) and academia (Southern University of Science and Technology, SUSTech and NYU), I am fascinated by rethinking system abstractions and optimizing their efficiency with full-stack approaches. The efforts, which led to publications at **SOSP'23**, **SIGMOD'25**, **PPoPP'25**, and **ICLR'25**, have strengthened my confidence to continue on this journey. Additionally, attending conferences such as VLDB, SIGMOD, and PPoPP further enabled me to stay abreast of trending breakthroughs. For these reasons I am determined to pursue a Ph.D. in computer systems.

The rapid growth of deep learning has fueled the success of transformative AI applications across domains. However, new ML algorithms are typically developed with little consideration for cost-effectiveness, making them impractical to be directly deployed in real production environments. Many attempts have been proposed to solve this, among which, three directions excite me the most: 1) develop new abstractions for emerging algorithmic patterns to enable holistic optimization opportunities; 2) apply fine-grained system techniques tailored to the algorithm and hardware characteristics; and 3) jointly consider different components in end-to-end ML pipelines for better resource utilization and scalability. Over the past few years, I have done research in several projects spanning these themes. For my Ph.D. study, I want to take a broader perspective, while still solving system challenges related to the above three aspects.

Research Experience at SUSTech & AWS Shanghai AI Lab

I had both my undergraduate and master's studies at DBGroup@SUSTech, advised by **Prof. Bo Tang** and **Prof. Xiao Yan**. My foundational experience was forged during my internship with the AWS Shanghai DGL team, working with **Dr. Minjie Wang**. I started the journey by working on optimizing **GNN training across diverse, large-scale scenarios**, tackling challenges from low-level system implementation to the algorithm layer. On the algorithmic side, I co-authored MuseGNN [4] (**ICLR 2025**), designing new GNN architectures that are both provably convergent and scalable to large graphs. This algorithm perspective guided my primary focus on systems research. During the exploration, I observed a surge in new GNN sampling algorithms that existing systems could not support, leading to ad-hoc implementation and inefficient sampling. As such, I identified a critical need for a new system abstraction to enable **efficient graph sampling on GPU**. I studied 15 popular graph sampling algorithms, summarized their patterns, and sketched all sampling procedures using unified matrix operations. This new abstraction allows for holistic system optimizations. I proposed IR-based GPU kernel fusion to eliminate redundant data materialization, sparse format selection for better operator efficiency, and irregular subgraph batching to improve GPU utilization. These combined efforts yielded up to 20x speedup in the sampling process and a 40% reduction in end-to-end GNN training. This work, gSampler [2], was accepted by **SOSP 2023**.

With a knowledge of GNN training and industry workloads, I then extended my focus to the **distributed training scenario**, leading a project that tackles parallelism and communication bottlenecks in collaboration with **Prof. James Cheng** at CUHK. Through extensive empirical evaluation, I observed that no single GNN parallelism (such as DistDGL [14] and P3 [1]) performs optimally under all configurations, each with its own trade-offs. By viewing GNN computation as sparse matrix multiplication, I summarized all four possible parallelism strategies that cut along different dimensions and designed a framework to adaptively determine the optimal parallelism at runtime. To implement a unified execution engine for different parallel schemes in the framework, I proposed to automatically insert GPU communication operators at model layer barriers while preserving compatible interfaces with DGL [11]. Moreover, for accurate parallelism selection, I built a profiling-based cost model to precisely navigate the complex trade-offs and adapt to different hardware and models. This auto-parallelism enabled an over 2x speedup compared to always using a static, single-strategy approach. These efforts led to my co-first-author publication, APT [5], at **PPoPP 2025**.

To further enable **efficient GNN training on super-large graphs** where the graph data cannot fit in CPU memory, I then led a project to address the I/O bottleneck by optimizing the underlying disk access pattern. Contrary to popular approaches such as Ginex [6] and MariusGNN [10] that suffer from either disk read amplification or degraded model accuracy, I proposed the **offline sampling scheme**, where all mini-batches are prepared before the model computation iterations. This new scheme allows us to continuously store node features of a batch on disk for better access locality, all while preserving the original training algorithm, thus ensuring no model accuracy degradation. To reduce disk usage due to data replication, I

proposed a novel disk cache with a node reordering method to reduce read amplification. This system, together with hierarchical feature storage across GPU and CPU memory and elaborate overlapping between computation and I/O, demonstrates that billion-scale graphs could be trained efficiently without costly distributed setups. This work, DiskGNN [9], was presented at **SIGMOD 2025** as my first-author publication.

Research Experience at NYU Systems Group

To broaden my expertise, I pivoted from GNNs to tackle core bottlenecks in **LLM and modern AI pipelines** during my internship at NYU with **Prof. Jinyang Li**. I started by building robust Retrieval-Augmented Generation (RAG) solutions. I first led the project, PolyG [8], aimed at improving **Graph-based RAG** which augments LLM generation using information retrieved from an underlying knowledge graph. After an extensive investigation, I observed that existing GraphQA benchmarks were biased towards one type of question focused on object discovery. As such, I curated a new GraphRAG benchmark, PolyBench, which covers a comprehensive set of question patterns derived from the triple format in knowledge graphs. Moreover, since current methods are not designed for solving diverse question types, I developed a novel GraphRAG solution that adaptively selects suitable graph traversal methods to retrieve high-quality context with 4x speedup on end-to-end serving latency. Both PolyG and PolyBench are fully open-source at <https://github.com/Liu-rj/PolyG>. I plan to submit this work to KDD 2026 as the first author.

While developing RAG pipelines, I noticed that the deployment of LLM applications has clear performance bottlenecks. I am currently working on a project to optimize **LLM serving latency in RAG scenarios** by reusing KV caches of an external knowledge base across different queries. Observing that existing systems such as CacheBlend [12] and Epic [3] suffer from heavy I/O overhead when loading KV caches from disk, I proposed a novel mixed-precision KV compression approach that eliminates the I/O bottleneck while maintaining LLM generation quality. Based on this new compression scheme, I am designing a tailored pipeline to overlap KV compression with model prefill and decode computation, aiming to further improve the end-to-end serving latency. I plan to submit this work to ICML 2026 as the first author.

Beyond system optimizations, I am also deeply interested in algorithmic developments. When building PolyG, it became clear that LLM-guided information retrieval is just one component of a truly intelligent agent, and that current agentic frameworks, such as ReAct [13] and Reflexion [7], face reliability issues under complex workflows. As such, I am now also working on **a new system abstraction for general LLM agents**, which interprets agentic execution using Intermediate Representation (IR) to improve robustness and explainability. The core principle is to propose a set of IR operators to express agent actions with a Data-flow Graph (DFG), where each IR operator has clear semantics and can be optimized independently.

Research Plan: In my future PhD study, I am eager to continue research on system and algorithm optimizations for AI pipelines, particularly for LLM agents. I believe that **versatile, efficient, and robust agent systems** are key to unlocking future AI developments. Specifically, 1) current LLM agents are intrinsically unstable, and thus new system abstractions for better agent program auto-tuning and reliable online execution are required; 2) agent programs are observed to have a rigorous development phase, where the agent’s capability are refined by interleaved evaluation and optimization, and thus this stage necessitates efficient and robust system support; 3) the deployment of agentic workflows consists of both LLM generation and external function calls, and thus joint optimizations on end-to-end resource utilization and scheduling are needed. This integrated approach—designing better abstractions and optimizing their execution workflows—is the line of research that I am most passionate about and wish to pursue in my PhD. In general, I am always excited to explore topics broadly in building cutting-edge AI infrastructure and applications.

Why University of Edinburgh: My long-term career objective is to lead a research group in academia or industry, pioneering the next generation of efficient and scalable AI infrastructure. University of Edinburgh is the ideal environment to achieve this, particularly due to the full-stack focus of the **MLsys** department. I am specifically drawn to work with **Prof. Luo Mai**, whose pioneering work on LLM serving and agents, such as ServerlessLLM and WaferLLM, aligns directly with my research vision. I aim to build the robust system support needed to make emerging approaches on agentic workflow optimization scalable in production. The leading research and inclusive environment make University of Edinburgh the perfect place for me to transition from a student researcher to a leader in the field.

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